

ABSTRACT

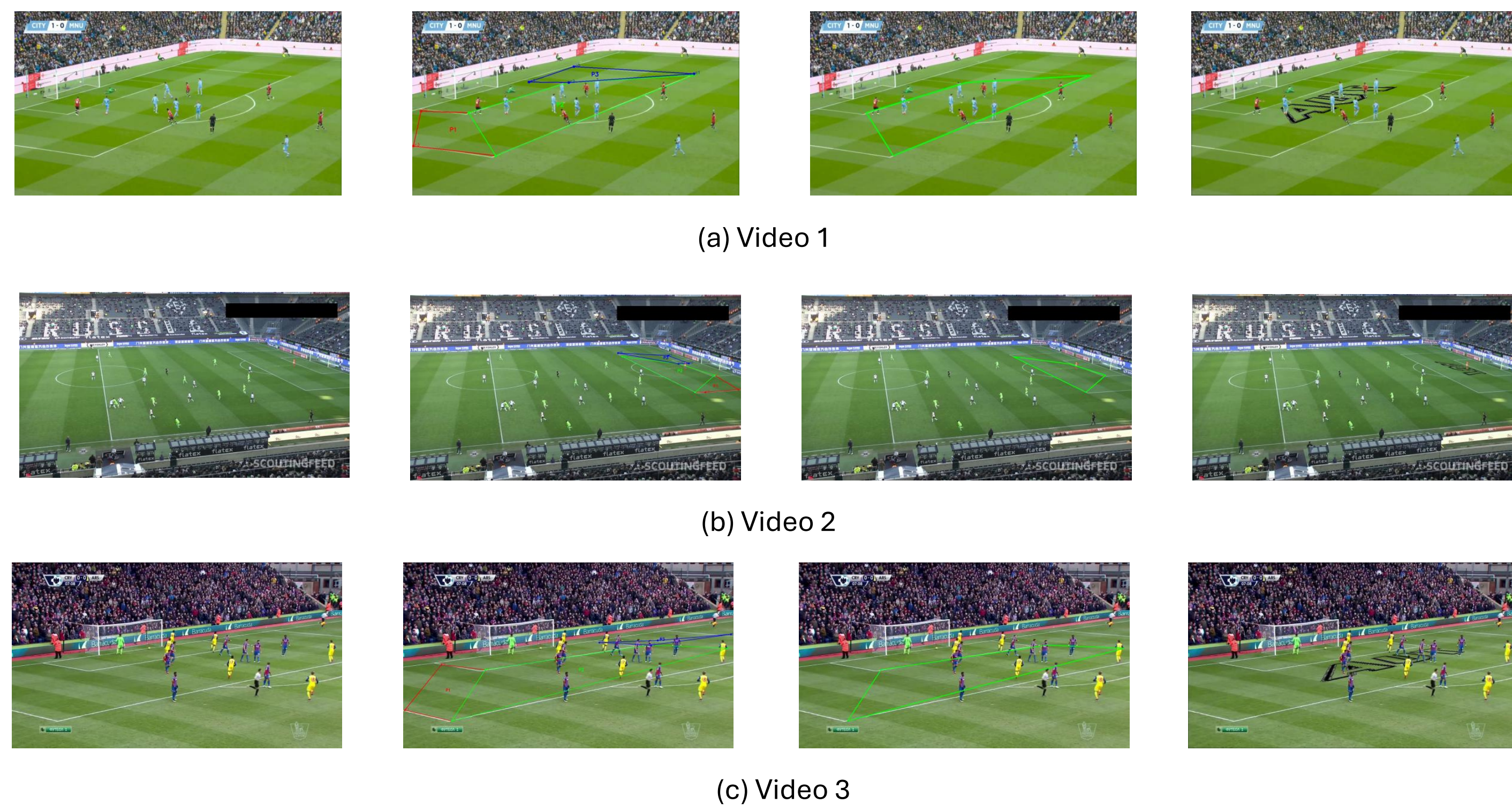
What do we solve? We address the problem of robust virtual advert placement in the soccer penalty area under occlusion and perspective distortion. We introduce an automatic procedure to select a geometrically consistent quadrilateral region from field calibration coordinates.

What did we find? Quantitative evaluations demonstrate that our method preserves high visual fidelity (SSIM of 0.97 and PSNR of 31.6 dB) while maintaining temporal consistency with flicker increase ($< 7\%$).

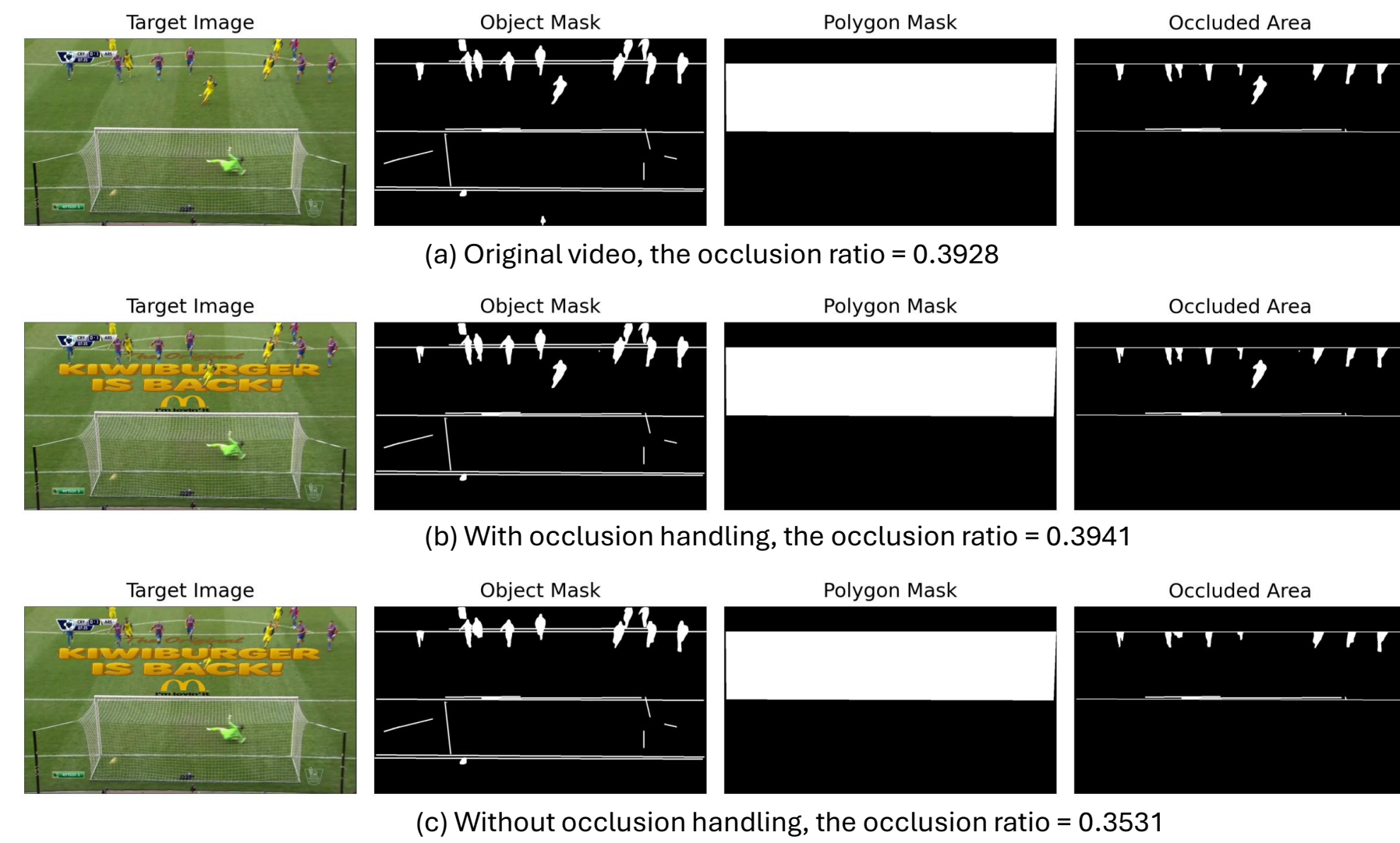
WHY DOES THIS MATTER?

- **High commercial demand with limited automation:** Virtual advertisement insertion in sports broadcasting is increasingly important for monetization, yet current workflows remain largely manual and labor-intensive.
- **Need for geometry- and occlusion-aware placement:** Effective advert placement requires accurate perspective alignment with the field and proper occlusion handling to ensure visual realism.

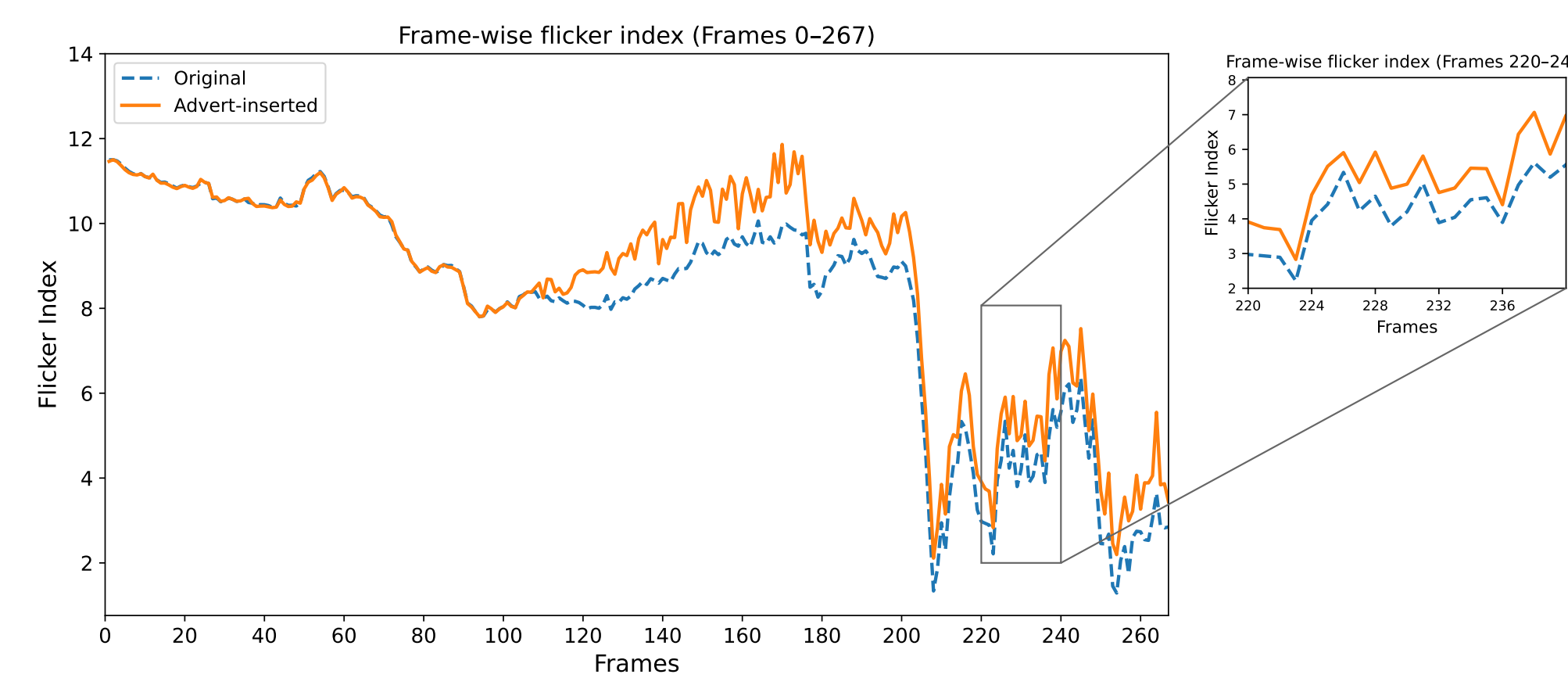
COMPARATIVE ANALYSIS



Qualitative Insights. The examples illustrates an automatic polygon selection to final advertisement placement, including occlusion handling is presented for representative video sequence.

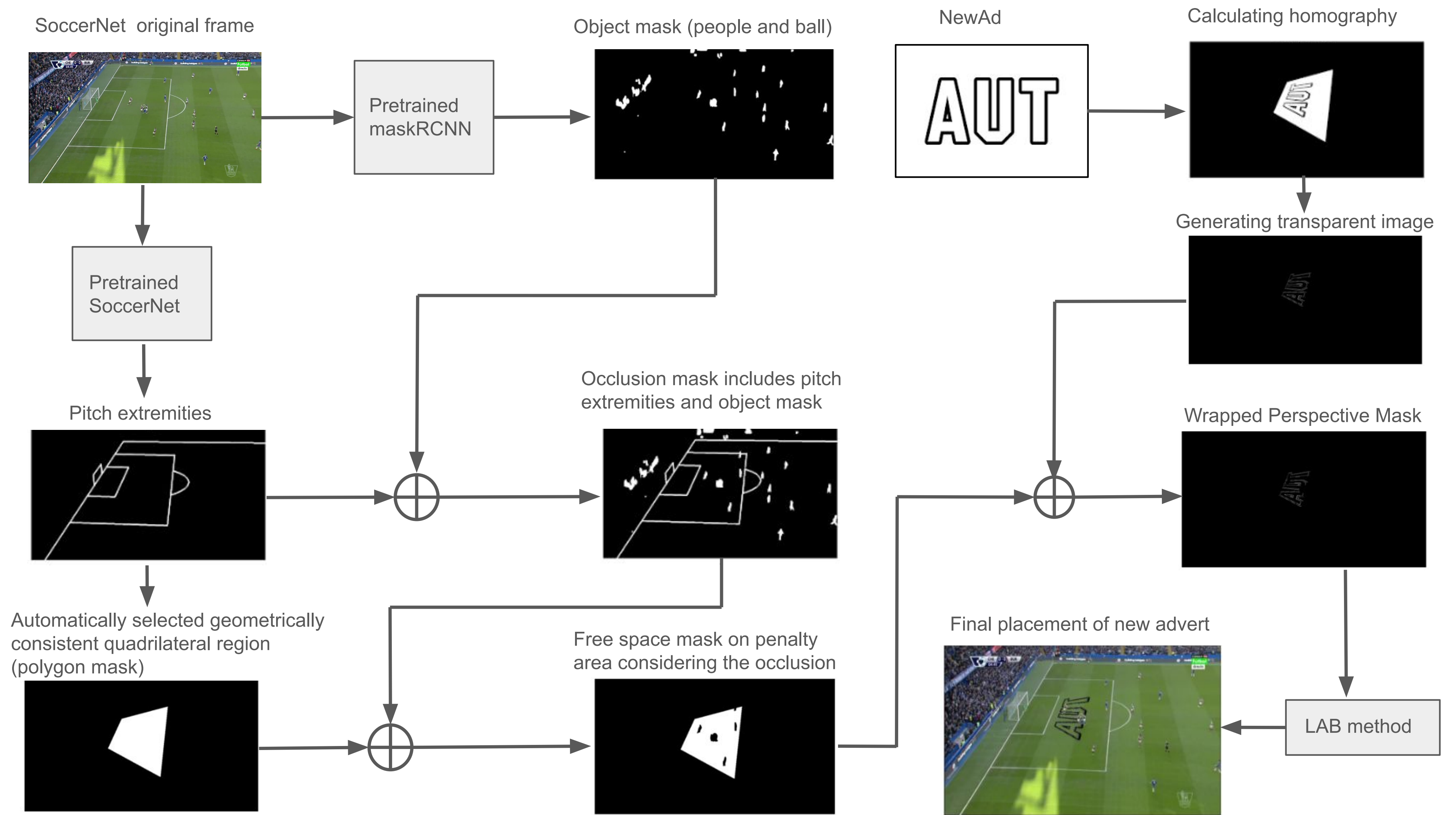


Visualization of the occluded area for representative Video.



The video achieve $< 7\%$ flicker index, confirming strong visual realism.

OUR APPROACH:



The approach integrates three components:

1. **Homography estimation** — A geometry-driven framework that automatically selects geometrically consistent quadrilateral region within the penalty area, eliminating the need for manual region annotation.
2. **Occlusion handling** — An occlusion-aware dynamic advert placement pipeline based on homography transformation and instance segmentation, enabling realistic advert integration.
3. **Advert warping and placement-** A method for seamless scene integration under varying lighting and perspective conditions within a pre-existing advert region and on static images.

CONCLUSION

Key outcomes: The proposed framework enables fully automated, occlusion-aware virtual advert placement in soccer broadcasts with high visual fidelity, achieving strong geometric alignment via homography-based warping on an automatically selected penalty-area quadrilateral while preserving realism.

Robustness: The method maintains correct occlusion relationships between players, the ball, and advertisements using Mask R-CNN-based instance segmentation and blending strategies, ensuring consistent integration in dynamic gameplay.

Impact: The approach provides a scalable solution for immersive sports broadcast augmentation and generalizes to other sports such as basketball and cricket.

REFERENCES

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- [2] Sukriti Dhang, Mimi Zhang, and Soumyabrata Dev. LaBINet-an approach for seamlessly integrating new advertisement into an existing scene. *IEEE Transactions on Artificial Intelligence*, 6(8):2281 – 2290, 2025.
- [3] Anthony Cioppa, Silvio Giancola, Adrien Deliege, Le Kang, Xin Zhou, Zhiyu Cheng, Bernard Ghanem, and Marc Van Droogenbroeck. Soccernet-tracking: Multiple object tracking dataset and benchmark in soccer videos. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3491–3502, 2022.